



Module 1

Introduction



Introduction



What is Machine Learning?

- the field of study that gives computers the ability to **learn** without being explicitly programmed (Arthur Samuel, 1959).

Label	Email
ham	Go until jurog point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...
ham	Ok lar... Joking wif u oni...
spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
ham	U dun say so early hor... U c already then say...
ham	Nah I don't think he goes to usf, he lives around here though
spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, Â£1.50 to rcv
ham	Even my brother is not like to speak with me. They treat me like aids patient.
ham	As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
spam	WINNER!! As a valued network customer you have been selected to receivea Â£900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030
ham	I'm gonna be home soon and i don't want to talk about this stuff anymore tonight, k? I've cried enough today.
spam	SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH11 and send to 87575. Cost 150p/day, 6days, 16+ TsandCs apply Reply HL 4 info
spam	URGENT! You have won a 1 week FREE membership in our Â£100,000 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C www.dbuk.net LCCLTD POBOX 4403LDNW1A7RW18
ham	I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing at all times.
ham	I HAVE A DATE ON SUNDAY WITH WILL!!
spam	XXXMobileMovieClub: To use your credit, click the WAP link in the next txt message or click here>> http://wap.xxxmobilemovieclub.com?n=QJKGIGHUJGCBL
ham	Oh k...i'm watching here;
ham	Eh u remember how 2 spell his name... Yes i did. He v naughty make until i v wet.
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ham	K tell me anything about you.
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	Pred. Spam	Pred. Ham
Spam		
Ham		

What is Machine Learning?

- A computer program is said to learn from **experience** E with respect to some **task** T and some **performance measure** P , if its performance on T , as measured by P , improves with experience E (Tom Mitchell, 1997).

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Pred. Spam

Pred. Ham

Spam

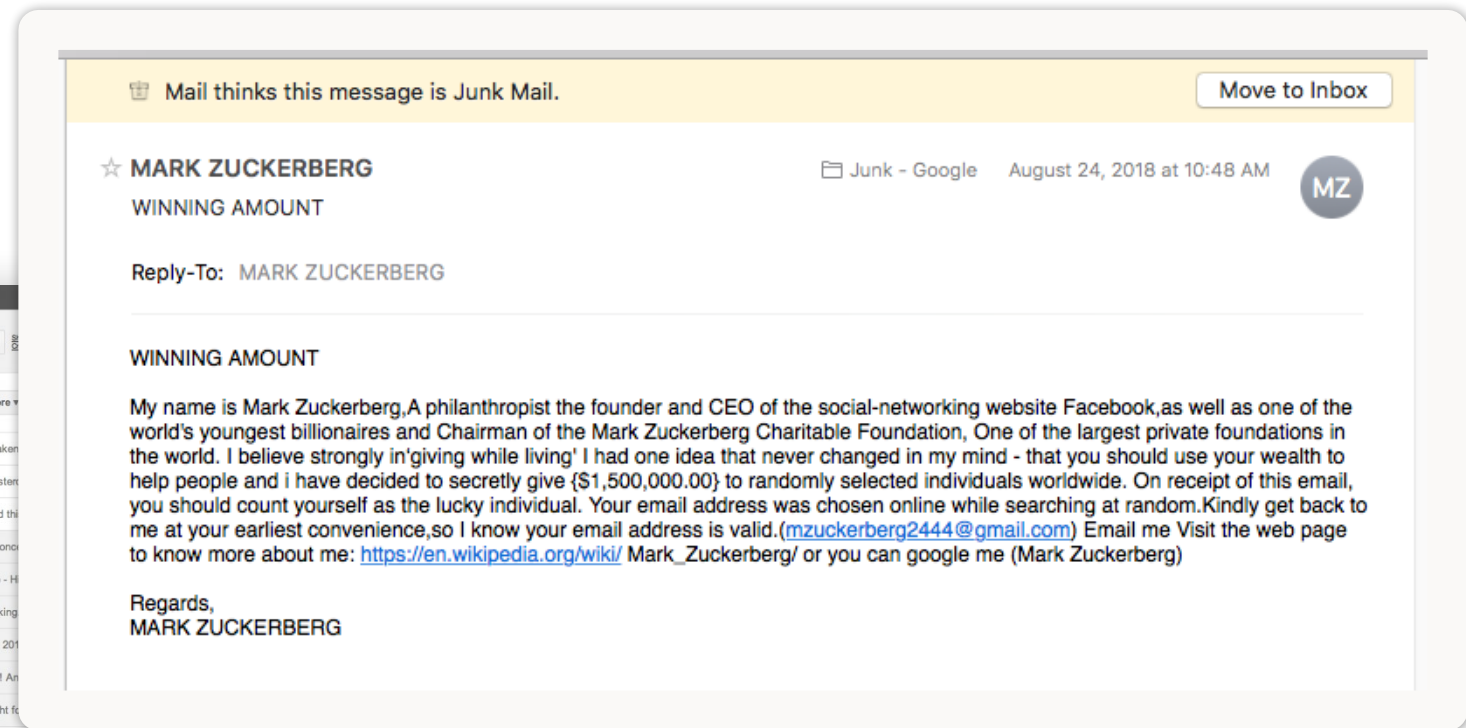
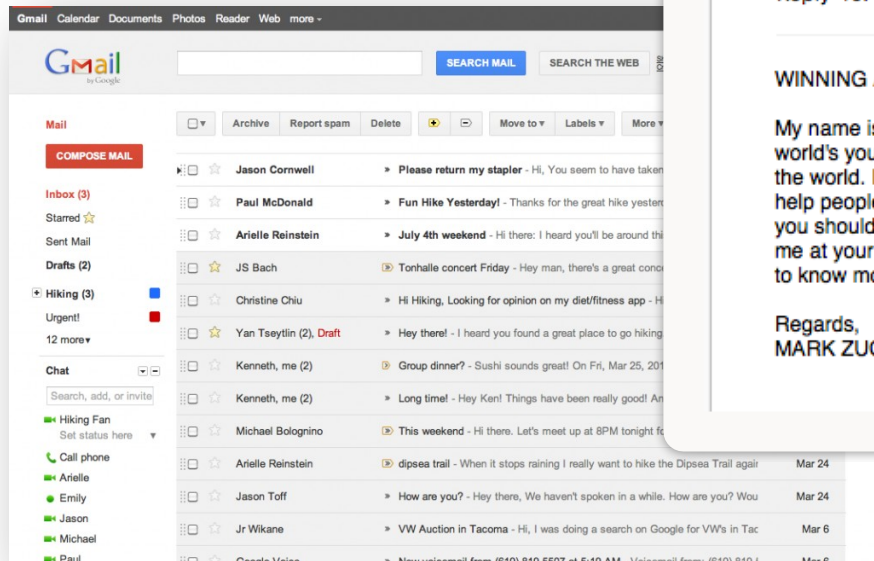


Ham



Task

□ Automatically classify an email as **Spam (Bad)** or **Ham (Good)**



Experience

□ a dataset with emails **labeled** Spam (Bad), Ham (Good)

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Performance Metrics

- a dataset with emails **labeled** Spam (Bad), Ham (Good)

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Scikit-Learn Classification Metrics

Scoring	Function	Comment
Classification		
'accuracy'	<code>metrics.accuracy_score</code>	
'balanced_accuracy'	<code>metrics.balanced_accuracy_score</code>	
'top_k_accuracy'	<code>metrics.top_k_accuracy_score</code>	
'average_precision'	<code>metrics.average_precision_score</code>	
'neg_brier_score'	<code>metrics.brier_score_loss</code>	
'f1'	<code>metrics.f1_score</code>	for binary targets
'f1_micro'	<code>metrics.f1_score</code>	micro-averaged
'f1_macro'	<code>metrics.f1_score</code>	macro-averaged
'f1_weighted'	<code>metrics.f1_score</code>	weighted average
'f1_samples'	<code>metrics.f1_score</code>	by multilabel sample
'neg_log_loss'	<code>metrics.log_loss</code>	requires <code>predict_proba</code> support
'precision' etc.	<code>metrics.precision_score</code>	suffixes apply as with 'f1'
'recall' etc.	<code>metrics.recall_score</code>	suffixes apply as with 'f1'
'jaccard' etc.	<code>metrics.jaccard_score</code>	suffixes apply as with 'f1'
'roc_auc'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovr'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovo'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovr_weighted'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovo_weighted'	<code>metrics.roc_auc_score</code>	

Types of ML Tasks

Supervised Learning

- Labeled data
- Direct feedback
- Predict outcome/future

Unsupervised Learning

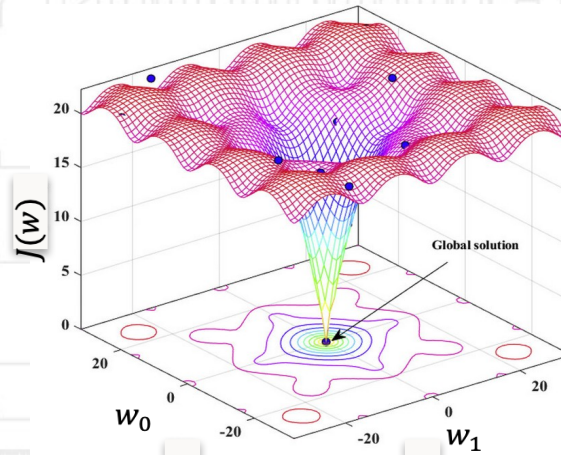
- No labels/targets
- No feedback
- Find hidden structure in data

Reinforcement Learning

- Decision process
- Reward system
- Learn series of actions



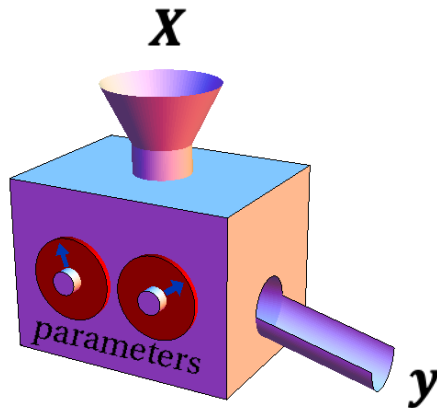
Loss Functions, Cost Functions



Supervised Learning

- Given a set of instances X
- There is an unknown function f that maps X to y

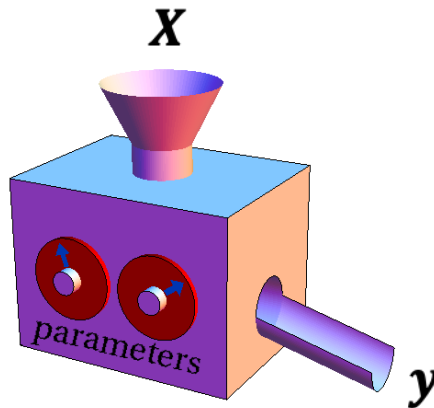
$f: X \xrightarrow{\text{maps}} y$



Supervised Learning

- Given a set of instances X
- There is an unknown function f that maps X to y

$f: X \xrightarrow{\text{maps}} y$



Date	Open	High	Low	Volume	Close
7/12/16	73.559998	73.750001	73.529999	12501200	73.209997
7/13/16	73.25	73.629997	72.940002	6399000	73.620003
7/14/16	73.790001	74.110001	73.610001	6646400	73.699997
7/15/16	73.980003	74.010002	73.440002	5994800	73.669998
7/18/16	73.639999	74.18	73.529999	5457300	73.839996
7/19/16	73.639999	73.959999	73.580002	4798100	73.660004
7/20/16	73.879997	74.110001	73.629997	5282000	73.790001
7/21/16	73.43	73.660004	73.139999	4391900	73.519997
7/22/16	73.760002	73.910004	73.419998	4316600	73.550003
7/25/16	73.550003	73.830002	73.410004	4734000	73.75
7/26/16	73.709999	74.07	73.410004	5248300	73.730003
7/27/16	73.660004	73.82	73.139999	5777900	73.32
7/28/16	73.139999	73.32	72.470001	4603200	73.239998
7/29/16	73.169998	73.25	72.760002	6302200	72.970001
8/1/16	72.809998	73.989998	72.809998	5868400	73.779999
8/2/16	73.699997	73.720001	72.629997	6513100	73.129997
8/3/16	73	73.400002	72.510002	6568100	72.940002
8/4/16	72.800003	73.489998	72.589996	4944900	73.300003
8/5/16	73.459999	73.949997	73.459999	5415400	73.760002
8/8/16	74	74.07	73.040001	6577200	73.339996
8/9/16	73.360001	73.879997	73.080002	6427900	73.540001
8/10/16	73.57	74.510002	73.389999	8052300	73.949997
8/11/16	74.5	74.800003	73.529999	6290300	73.800003
8/12/16	73.800003	74.120003	73.559998	9994200	73.889999
8/15/16	73.809998	74.010002	73.110001	9636100	73.32
8/16/16	72.589996	73.309998	72.209999	9763300	72.889999
8/17/16	71.989998	72.949997	71.989998	14343300	72.93

$h: X \xrightarrow{\text{maps}} y$

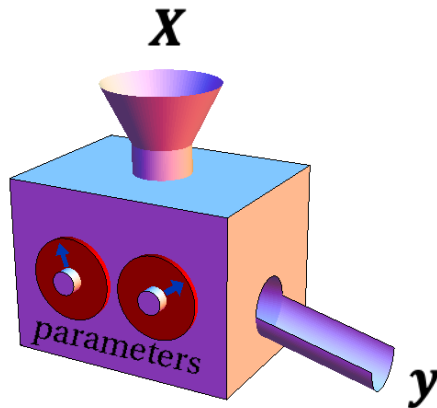


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Supervised Learning

- Given a set of instances X
- There is an unknown function f that maps X to y

□ $f: X \xrightarrow{\text{maps}} y$



hours_per_week	income	probability_over_50k
40	0	0.08284
60	1	0.66685
35	1	0.17607
60	1	0.54052
40	0	0.06032
...	...	...
40	0	0.11748
56	1	0.99997
60	1	0.86580
14	0	0.00699
45	0	0.16069

□ $h: X \xrightarrow{\text{maps}} y$



What is a Loss Function?

- ❑ **Distance** between ML algorithm output and the expected output.
- ❑ Calculates the error of the hypothesis.
- ❑ Generally referred to an instance for a single training sample.

Regression

$$(y_i - h_{\mathbf{w}}(x_i))^2$$
$$|y_i - h_{\mathbf{w}}(x_i)|$$

Classification

$$-(y_i \log(h_{\mathbf{w}}(x_i)) + (1 - y_i) \log(1 - h_{\mathbf{w}}(x_i)))$$

What is a Cost Function?

- ▣ Sum (or average) of the loss function over all observations
- ▣ Used by ML algorithms to obtain estimates $\mathbf{w}$

$$J(w_0, w_1, \dots, w_k) = \frac{1}{n} \sum_{i=1}^n (y_i - h_w(x_i))^2$$

$$J(\mathbf{w}) = \frac{1}{n} \|\mathbf{y} - h_w(\mathbf{X})\|_2^2$$

What is a Cost Function?

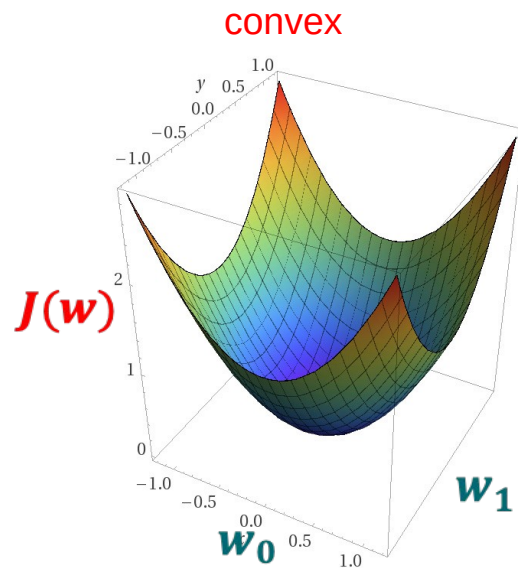
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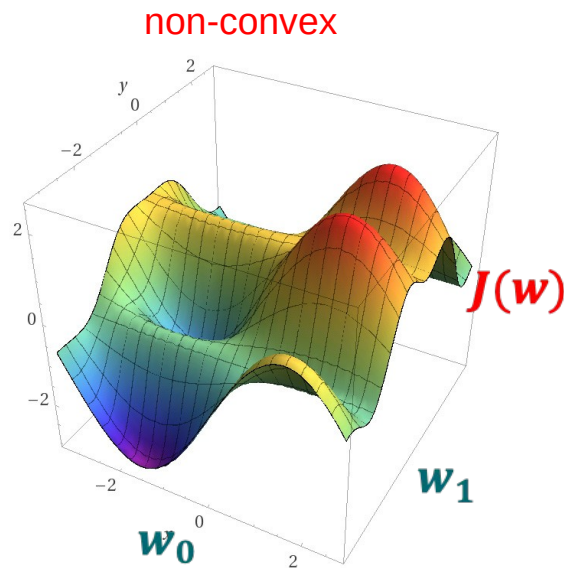
$$J(\mathbf{w})$$

Convex Cost Functions

- ❑ Convex functions have a unique global minimum and no local minima
- ❑ Algorithms with convex cost functions have simpler optimizations.



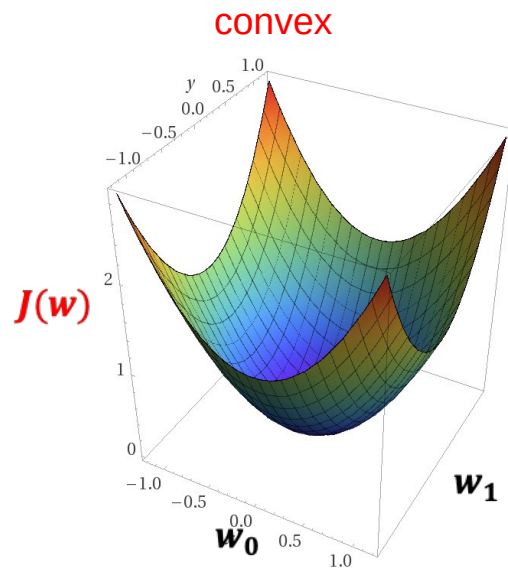
Computed by Wolfram|Alpha



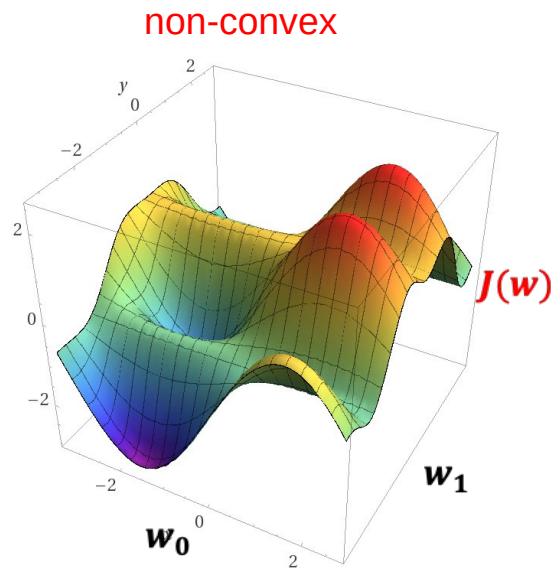
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Convex Cost Functions

- ❑ Non-convex cost function have both local and global minima
- ❑ Non-convex cost functions require advanced optimization to find the global minimum. Often use gradient descent.



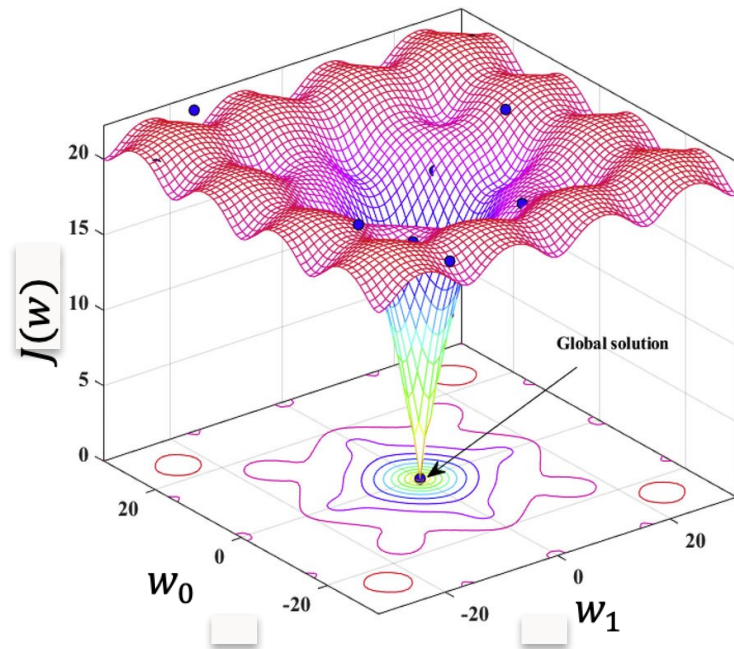
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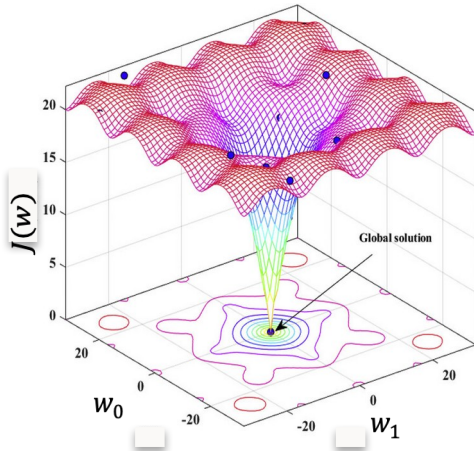


What is an Objective Function?

- ❑ Instructions we give the algorithm to select parameters $\mathbf{w}$.

$$\min_{\mathbf{w}} J(\mathbf{w})$$

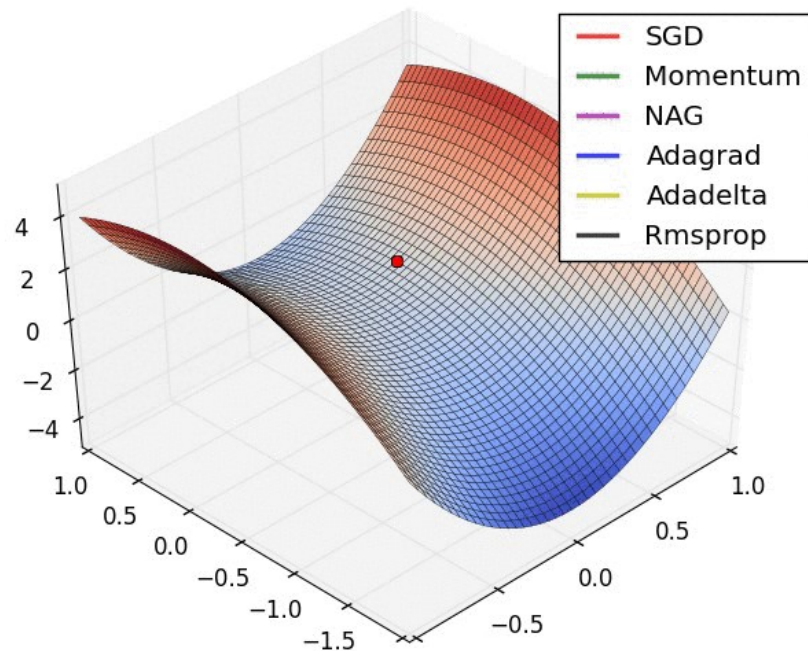
- ❑ In other words, select $\mathbf{w}$ such that the cost function J is a minimum



What is an Objective Function?

- Optimizers allow us to fine-tune parameters during training.

$$\min_w J(w)$$



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What is the most common notation for the COST function for an algorithm?

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In linear regression the residuals sum of squares (RSS) is the

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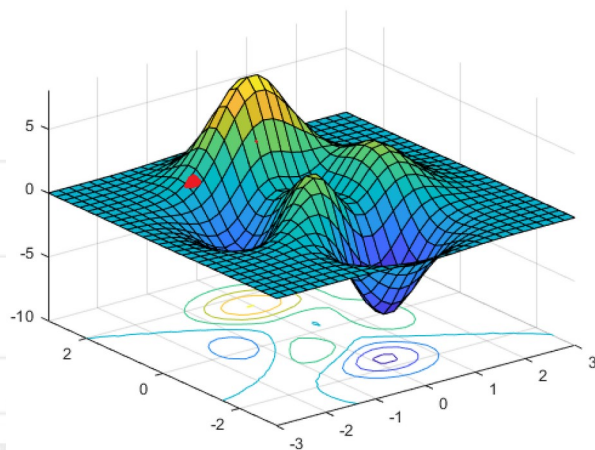


Which of the following best describes a loss function in machine learning?

☐ Start presenting to display the poll results on this slide.



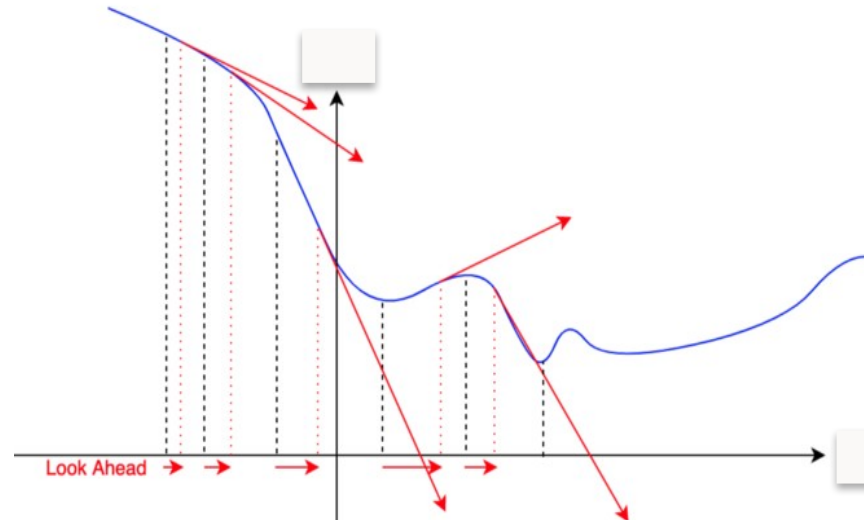
Gradients



What is the Gradient to a Cost Function?

- ❑ The gradient to a cost function refers to the instantaneous rate of change (descent).
- ❑ Mathematically, it is derivative of the cost function with respect to the estimated parameters

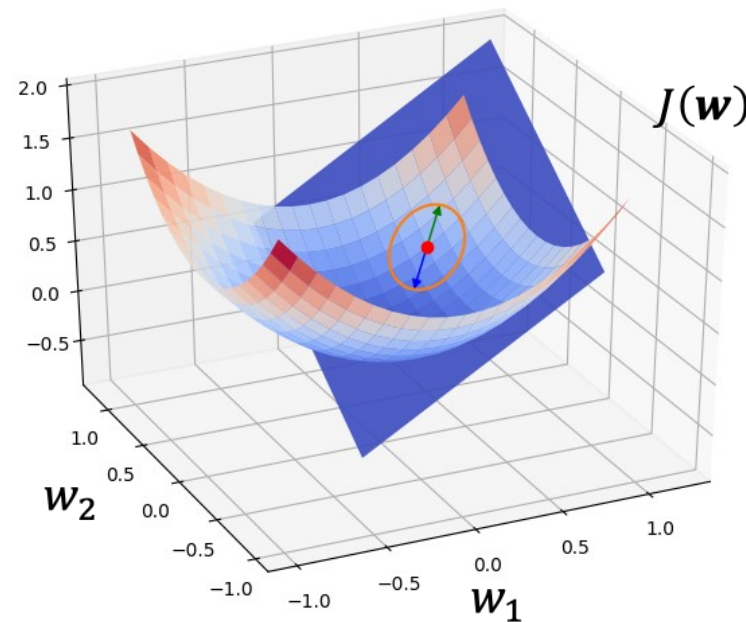
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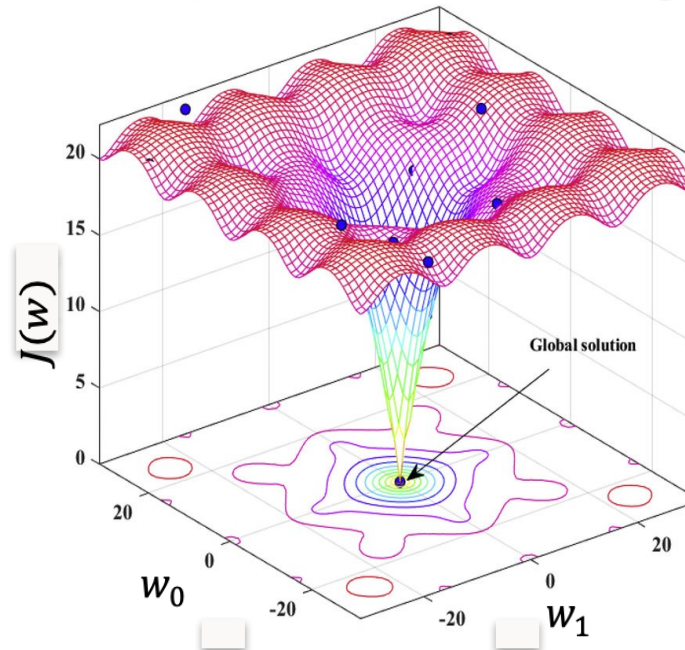
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What is the Gradient to a Cost Function?

- ❑ Optimizers search for local and global minima
- ❑ Local and global minima are found on the values of $\mathbf{w}$ such that the gradient is zero
- ❑ An important job of the optimizers is to get "unstuck" of local minima

$$\nabla J(\mathbf{w}) = \frac{\delta J(\mathbf{w})}{\delta \mathbf{w}}$$



Metrics vs Cost Functions

- ❑ Metrics are used to check the progress of the ML algorithm.
- ❑ Cost functions are used in the inner workings of the algorithm to update $\mathbf{w}$ and may NOT be known by practitioner.

Regression	
'explained_variance'	<code>metrics.explained_variance_score</code>
'max_error'	<code>metrics.max_error</code>
'neg_mean_absolute_error'	<code>metrics.mean_absolute_error</code>
'neg_mean_squared_error'	<code>metrics.mean_squared_error</code>
'neg_root_mean_squared_error'	<code>metrics.mean_squared_error</code>
'neg_mean_squared_log_error'	<code>metrics.mean_squared_log_error</code>
'neg_median_absolute_error'	<code>metrics.median_absolute_error</code>
'r2'	<code>metrics.r2_score</code>
'neg_mean_poisson_deviance'	<code>metrics.mean_poisson_deviance</code>
'neg_mean_gamma_deviance'	<code>metrics.mean_gamma_deviance</code>
'neg_mean_absolute_percentage_error'	<code>metrics.mean_absolute_percentage_error</code>
'd2_absolute_error_score'	<code>metrics.d2_absolute_error_score</code>
'd2_pinball_score'	<code>metrics.d2_pinball_score</code>
'd2_tweedie_score'	<code>metrics.d2_tweedie_score</code>

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The derivative of the cost function with respect to the estimated parameters is called:

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A minimum of the cost function occurs when the gradient is equal to zero.

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Local and global minima of a cost function are found on the values of w such that the gradient $J(w)$ is zero

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An important job of the optimizers is to get "unstuck" of the global minimum

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Metrics are the functions used to update the parameters w .

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Python